PRO2000 - PROGRAMMING

Week 41- Embedded C++ Programming

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Today's topics

- Embedded Programming and Microcontrollers
- Arduino
- Coding in Arduino IDE
- Coding in C++

Embedded Programming

- Embedded programming is the process of writing software, often called firmware, to control specialized hardware devices.
- Characteristics: special purpose, resource limited, direct hardware control, real-time demands
- Platforms/microcontrollers: like ATmega328P of Arduino Uno, STM32, ESP32, etc.
- Languages: Low/mid-level languages like C, C++ and Assembly.

ARDUINO

- Arduino is the go-to gear for artists, hobbyists, students, and anyone with a gadgetry dream.

- Arduino rose out of another formidable challenge: how to teach students to create electronics, fast.

- With Arduino, you can control almost everything around you be it simple LED or giant Robots.

ARDUINO

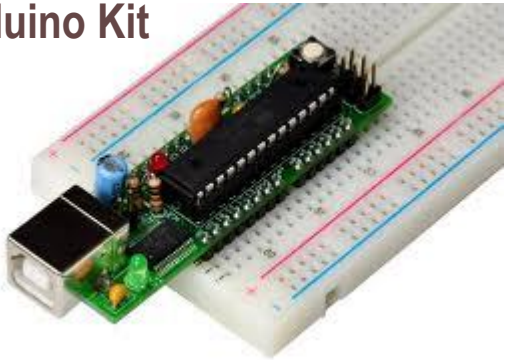
- Basically Arduino is Microcontroller.

- Microcontroller is microprocessor with memory, RAM and some other peripheral connected with it.

- The Arduino Uno is a microcontroller board based on the ATmega328 .
  The ATmega328 has Flash memory of 32 KB (with 0.5 KB used for the bootloader). It also has 2 KB of SRAM and 1 KB of EEPROM

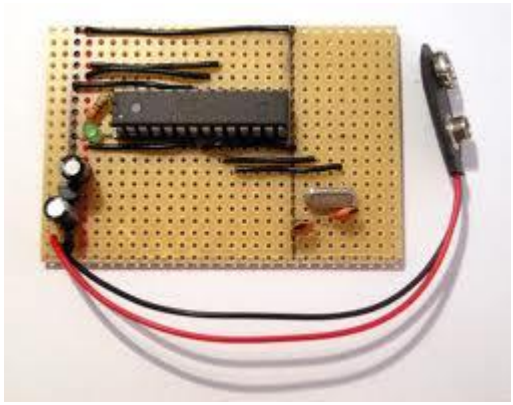
DIFFERENT TYPES OF ARDUINO

Boarduino Kit



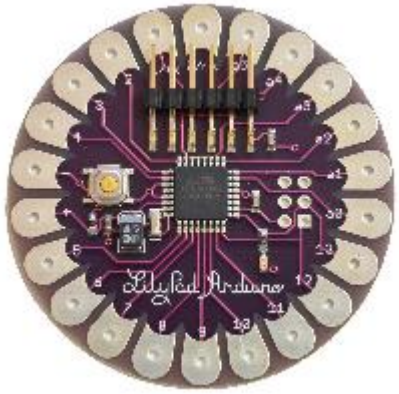


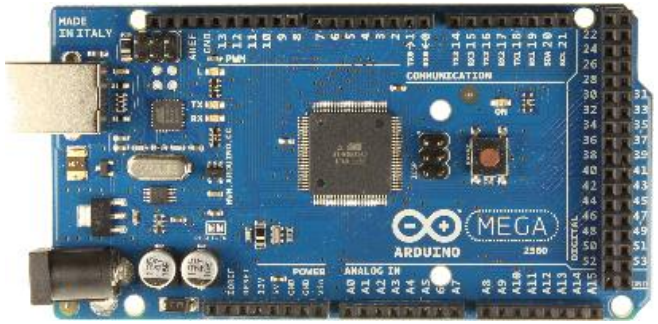
Arduino Uno



DIY Arduino

Arduino LilyPad

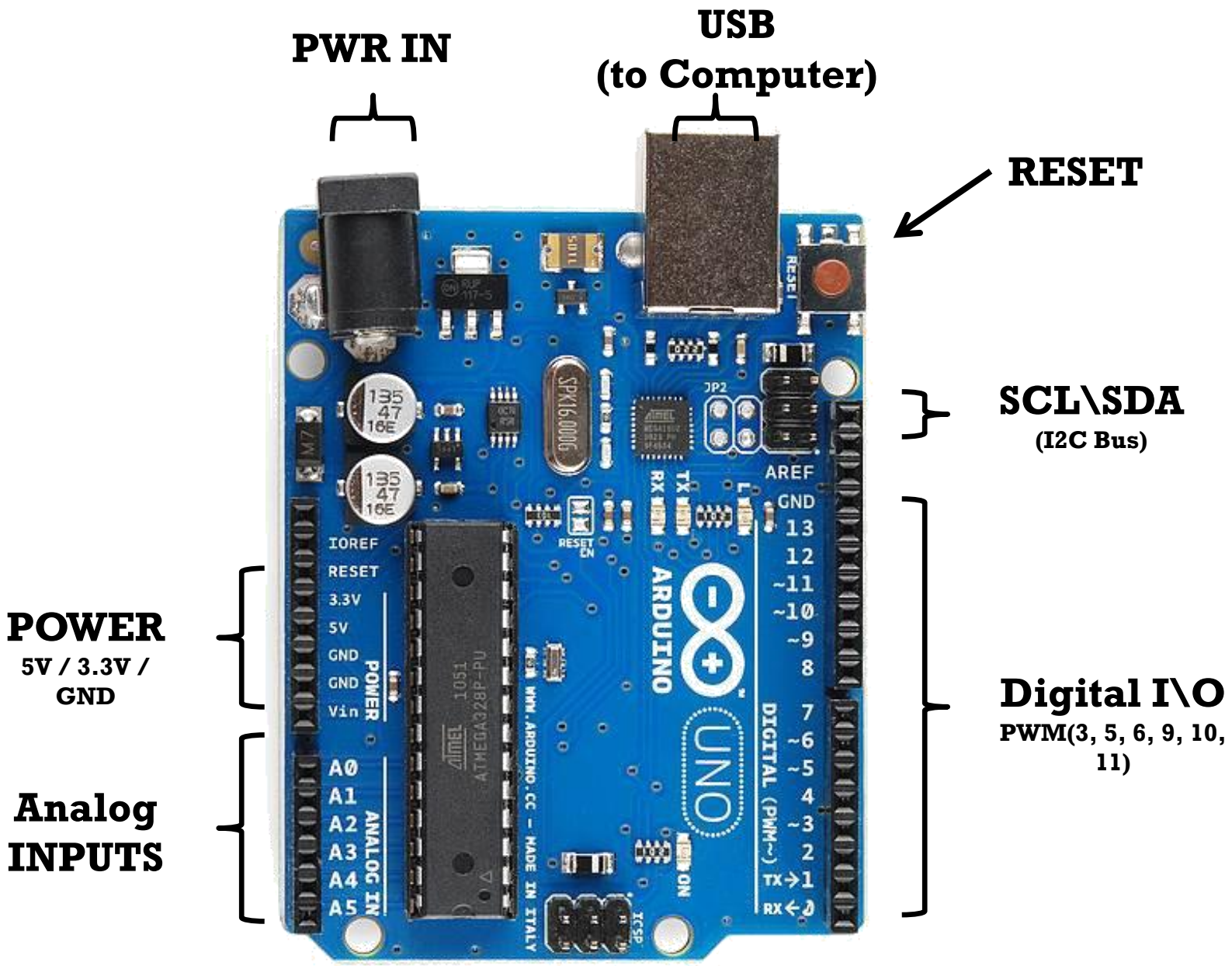




Arduino Mega 2560

ARDUINO UNO

- What does it have?
  - 14 Digital In/Out pins (6 can be used as PWM)
  - 6 Analog Inputs
  - A USB Connection
  - A Power Jack
  - Reset Button
  - On-board LED
  - SCL/SDA pins (Serial Clock/ Serial Data pins)
- In short, it contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.


PWR IN
USB
(to Computer)
RESET
SCL\SDA
(I2C Bus)
POWER
5V / 3.3V /
GND
Analog
INPUTS
Digital I\O
PWM(3, 5, 6, 9, 10,
11)
IOREF
RESET
3.3V
5V
GND
GND
Vin
POWER
A0
A1
A2
A3
A4
A5
ANALOG IN
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM~)
ARDUINO
UNO
ATMEGA328P-PU
WWW.ARDUINO.CC – MADE IN ITALY
ICSP
ON
JP2
SPX16.000G

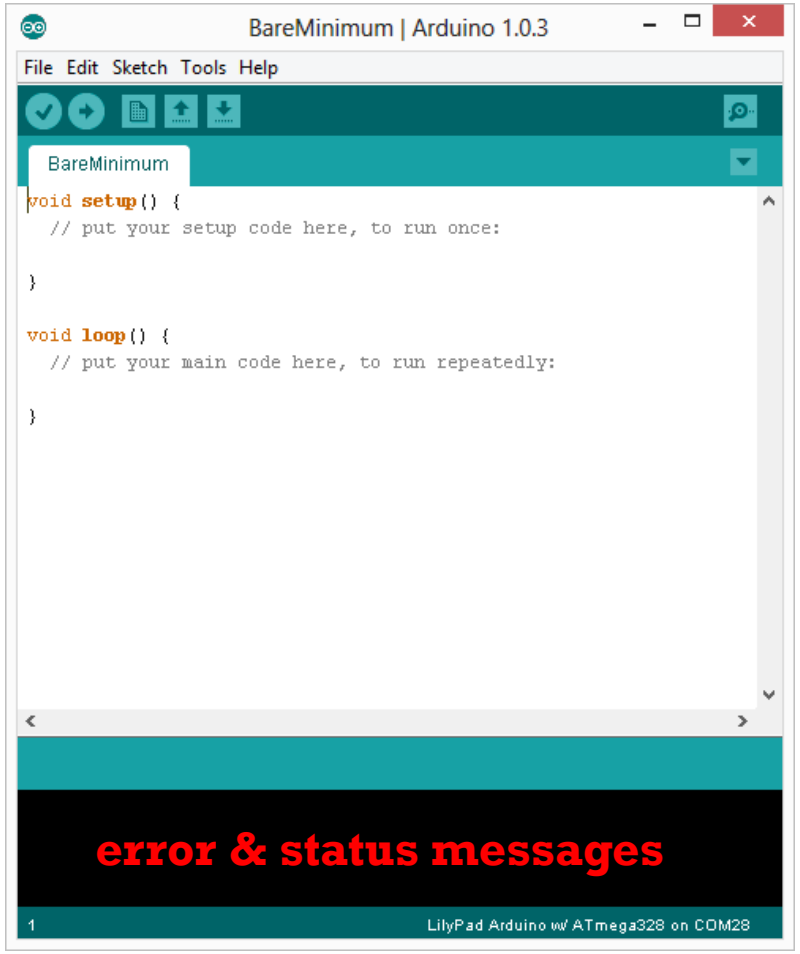
HOW TO CODE IN ARDUINO

- You need to download Arduino IDE (Integrated Development Environment).
- Arduino IDE is available for all Mac, Windows, and Linux.



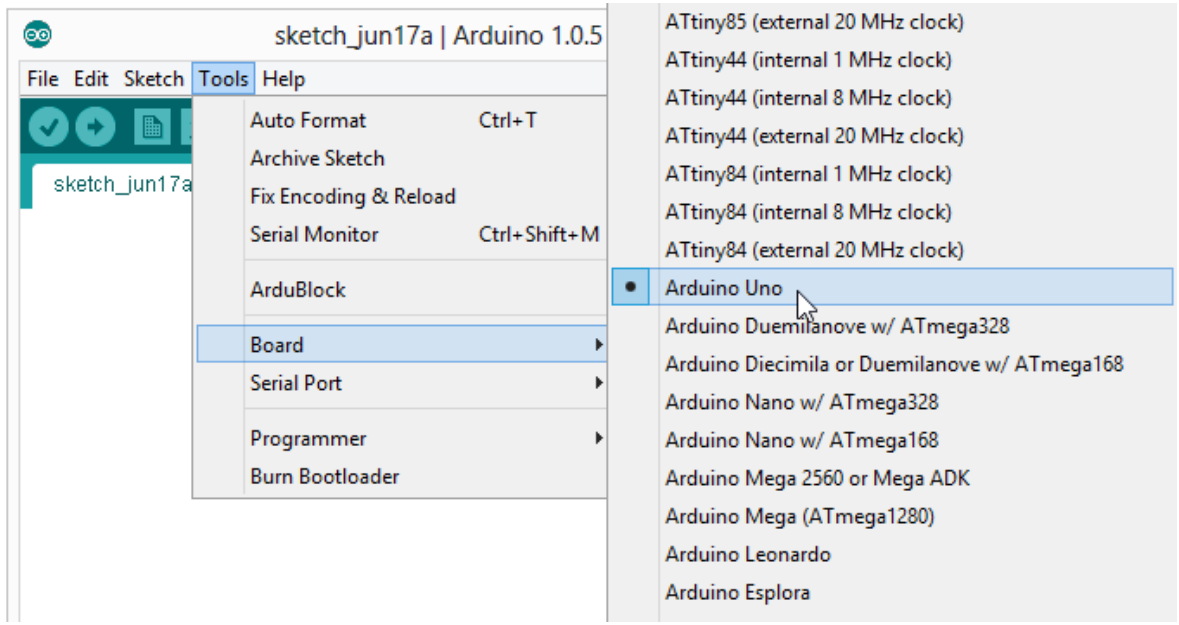
HOW TO CODE IN ARDUINO …

- Once you have downloaded and installed/extracted the folder, you can directly run Arduino.exe, which will take you to its IDE.
- The IDE will look like the shown screenshot.



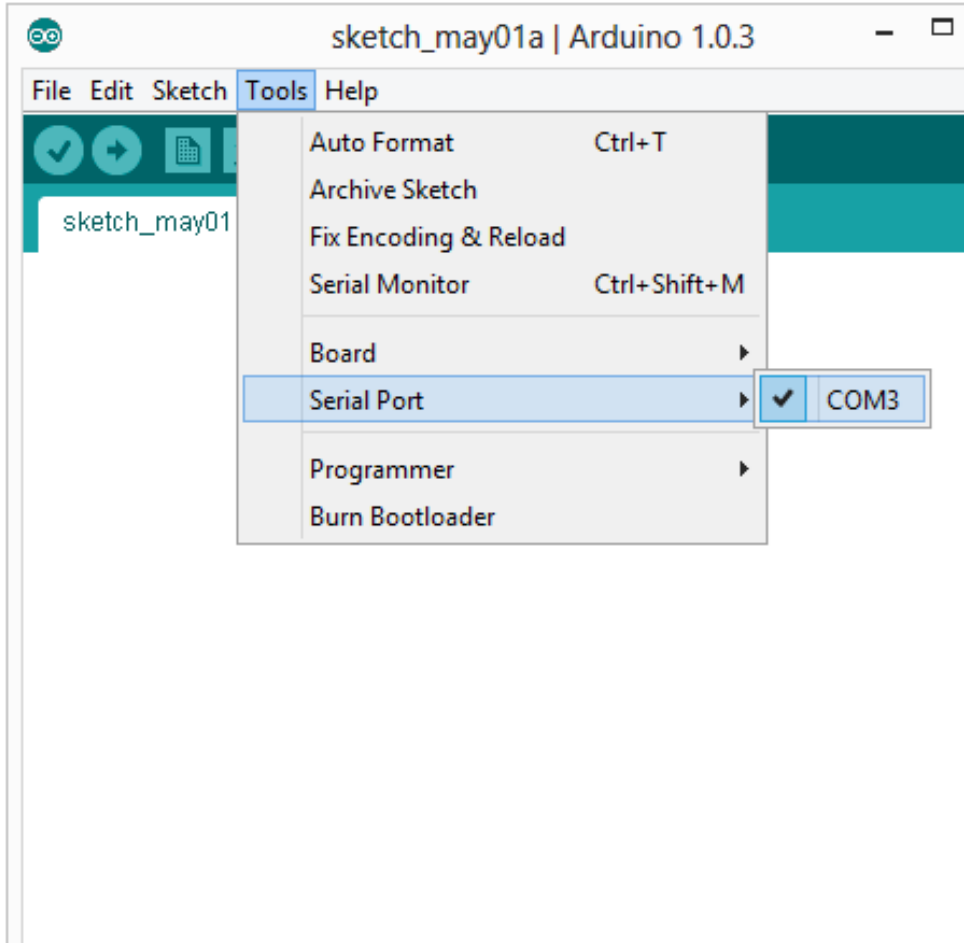
PROGRAM YOUR ARDUINO

- Before you start programming, double check that correct board is selected under Tools → Board.
- Now, you can start playing with Arduino.



PROGRAM YOUR ARDUINO

- The Arduino Uno can be programmed with the Arduino software. Select "Arduino Uno from the Tools > Board menu (according to the microcontroller on your board).
- All the peripheral connected with Computers are using Serial Port.
- You can check port for Arduino Uno in Device Manger.



INPUT VS OUTPUT

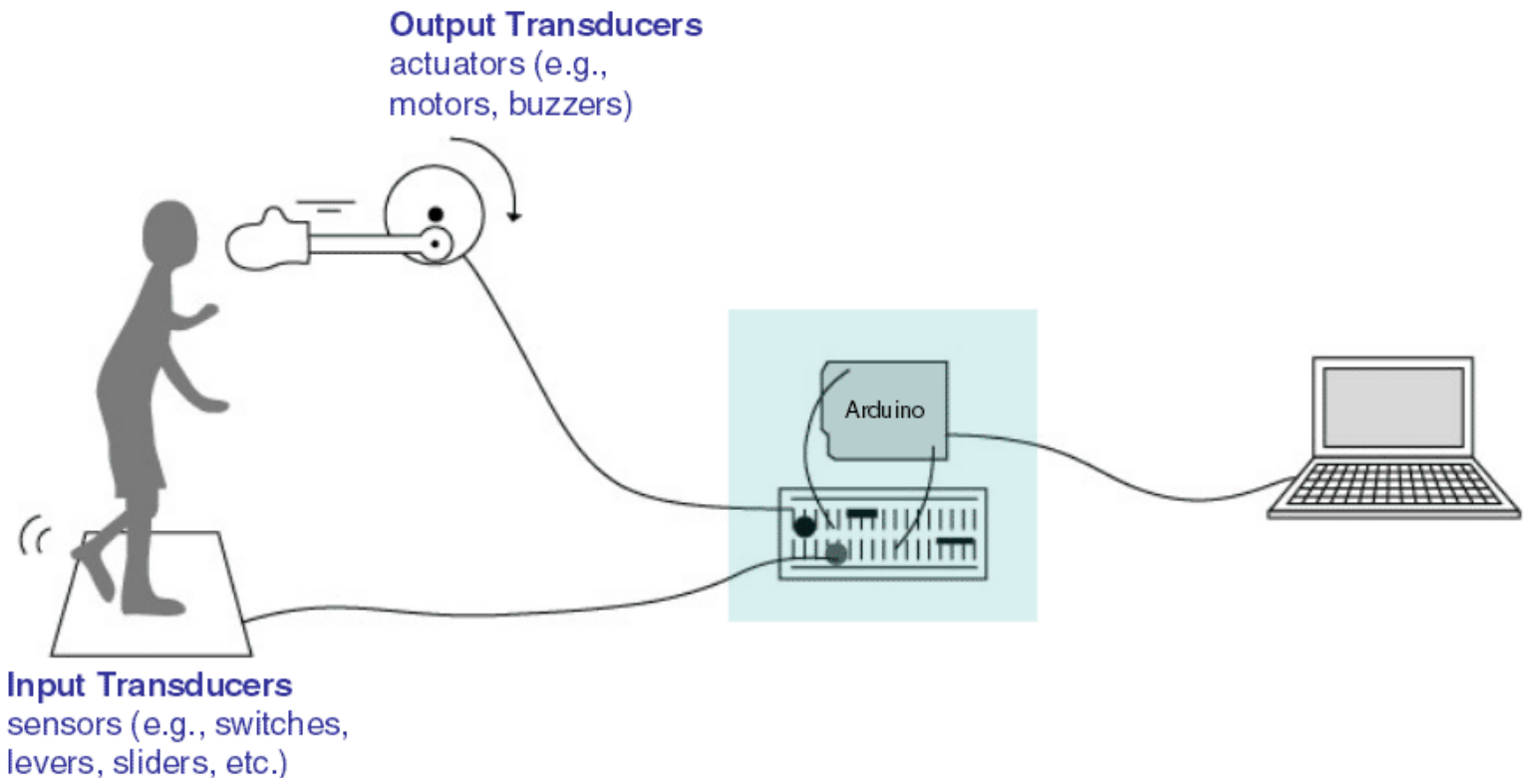


Image from *Theory and Practice of Tangible User Interfaces* at UC Berkley

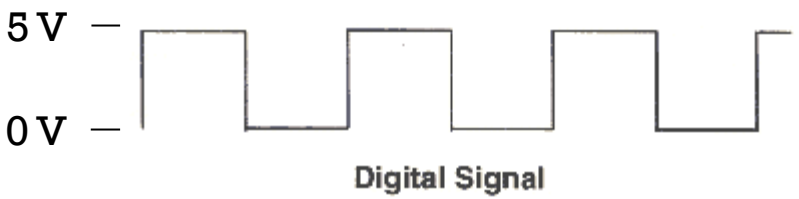
6 MAJOR CONCEPTS

- digitalWrite()
- analogWrite()
- digitalRead()
- If (statements) / Boolean
- analogRead
- Serial Communication

ANALOG VS DIGITAL

- Microcontrollers are **digital** devices – ON or OFF. Also called – discrete.

- **Analog** signals are anything that can be a full range of values.





ANALOG VS DIGITAL

- Analog Sensors

Sensors	Variables
Mic	soundVolume
Photoresistor	lightLevel
Potentiometer	dialPosition
Temp Sensor	temperature
Flex Sensor	bend
Accelerometer	tilt/acceleration

- Digital Sensors
  - Digital sensors are more straight forward than Analog.
  - No matter what the sensor there are only two settings: On and Off

- Example, Push button, Switch

SERIAL COMMUNICATION

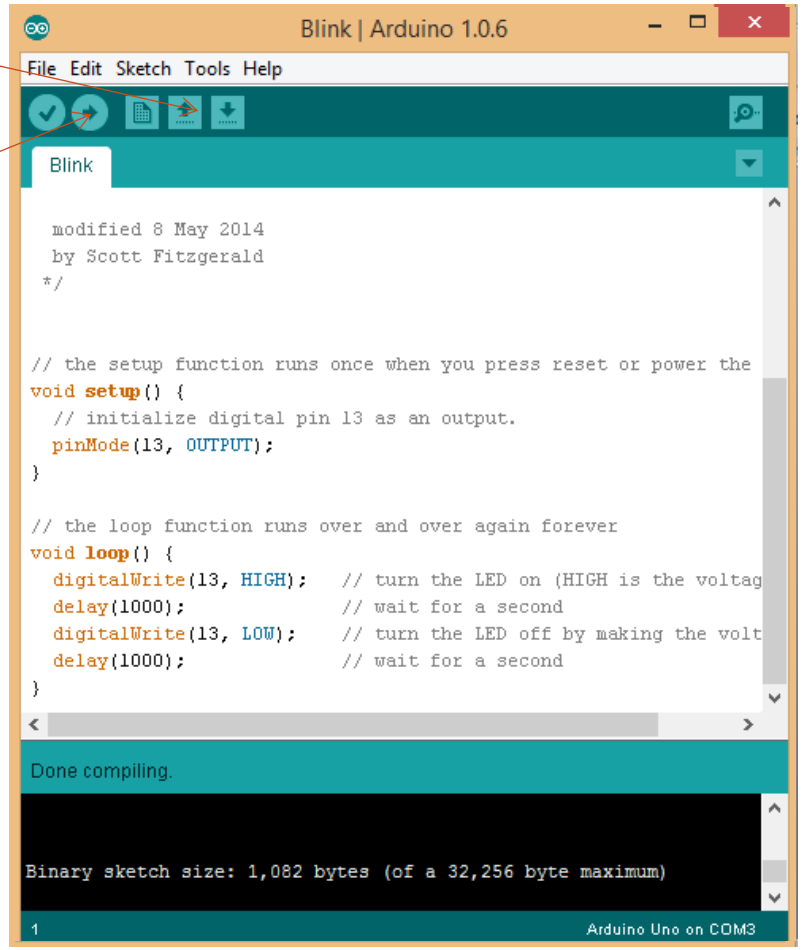
- "Serial" because data is broken into bits, each sent one after another in a single wire.
- Compiling turns your program into binary data (ones and zeros)
- Uploading sends the bits through USB cable to the Arduino
- The two LEDs near the USB connector blink when data is transmitted.
  - RX blinks when the Arduino is receiving data.
  - TX blinks when the Arduino is transmitting data

PROJECT #1 LED BLINK DIGITALWRITE()

Upload

Compile

Status Message



https://docs.arduino.cc/language-reference/en/functions/digital-io/digitalwrite/

HOW TO CODE IN C++

- Embedded programming in C++ allows to write clean, object-oriented, type-safe code without adding runtime or memory overhead.
- In resource constrained environments, C++ uses compile time computations like constexpr, and strict type like enum.
- Let's target the ATmega328P microcontroller of Arduino Uno.

HOW TO CODE IN C++ …

- Embedded C++ example to blink the built-in LED (Pin 13) on an Arduino Uno.
- On the Uno, digital Pin 13 corresponds to Port B, Pin 5.
- We manipulate it using three hardware registers: DDRB (Direction), PORTB (Data Output), and PINB (Input/Toggle).

HOW TO CODE IN C++ …

- Libraries- AVR delay utility

```
#include <stdint.h>
#include <util/delay.h>
```

- Strongly typed enum for safety

```
enum class PinState : uint8_t {    Low  = 0,    High = 1};
```

- Hardware mapping for ATmega328P Port B

```
struct PortB_Registers {
        volatile uint8_t PIN; // Input Pins Address (0x23)
        volatile uint8_t DDR; // Data Direction Register (0x24)
        volatile uint8_t PORT; // Data Register (0x25)
};
```

HOW TO CODE IN C++ …

- Zero-Overhead C++ Driver Class

```
class UnoLed {
private:
   // Base address for Port B registers on ATmega328P
   static constexpr uintptr_t PORTB_BASE = 0x23;

   // Hardcoded bitmask for Pin 13 (Bit 5 of Port B)
   static constexpr uint8_t LED_PIN_MASK = (1 << 5);

   // Cast the raw memory address to our register structure
   inline PortB_Registers* regs() const {
      return reinterpret_cast<PortB_Registers*>(PORTB_BASE);
   }
```

```
public:
   // Standard constructor
   constexpr UnoLed() = default;

   // Configure Port B Pin 5 as an Output
   void init() {
      regs()->DDR |= LED_PIN_MASK;
   }
   // Set LED explicitly On or Off
   void write(PinState state) {
      if (state == PinState::High) {
         regs()->PORT |= LED_PIN_MASK;  // Set bit high
      } else {
         regs()->PORT &= ~LED_PIN_MASK; // Clear bit low
      }
   }
   // Hardware-accelerated toggle (Writing a '1' to the PIN register toggles an AVR output pin instantly)
   void toggle() {
      regs()->PIN = LED_PIN_MASK;
   }
};
```

HOW TO CODE IN C++ …

- Main Application Entry Point

```
int main() {
  UnoLed led;
  led.init(); // Setup Pin 13 as output

  while (true) {
    // Method A: Explicit On/Off using Enums
    led.write(PinState::High);
    _delay_ms(500); // 500 millisecond delay

    led.write(PinState::Low);
    _delay_ms(500);

    // Method B: Ultra-fast toggle alternative
    // led.toggle();
    // _delay_ms(500);
  }

  return 0;
}
```

- In C++, it completely replaces the Arduino setup() and loop() structure with a standard C++ main() loop,
  - It saves valuable flash memory on framework initializations.
- To compile and run:
  - Download, install, and configure the Arduino IDE
  - Write C++ code
  - Load→ Observe by changing the arguments of the delay function.

HOW TO CODE IN C++ …

- Automate it in VS Code (**tasks.json**)

```
{
   "version": "2.0.0",
   "tasks": [
      {
         "label": "Build and Flash Uno",
         "type": "shell",
           "command": "avr-g++ -c -g -Os -Wall -fno-exceptions -ffunction-sections -fdata-sections -
           mmcu=atmega328p -DF_CPU=16000000L main.cpp -o main.o
           && avr-g++ -g -Wall -Os -mmcu=atmega328p -Wl,--gc-sections main.o -o main.elf
           && avr-objcopy -O ihex -R .eeprom main.elf main.hex
           && avrdude -F -V -c arduino -p m328p -P COM3 -b 115200 -U flash:w:main.hex:i",
         "group": {
            "kind": "build",
            "isDefault": true
         },
         "problemMatcher": []
      }
   ]
}
```

Up next: Exercises

- Now: 2 hours (12:15 to 14:00) of exercises (found on Canvas).

- Short exercise summary:
  - *Write, compile, and debug embedded programs with emphasis on Arduino IDE and C++ with VS Code.*
- Remember:
  - You should do approximately 4 more hours of the course (in a study group or alone) *before* our next session together. :-)